

Codes and Iterative Decoding on General Graphs

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I. INTRODUCTION

Until recently, most known decoding procedures for error-correcting codes were based either on algebraically *calculating* the error pattern or on some sort of *tree or trellis search*. With the advent of turbo coding [1], a third decoding principle has finally had its breakthrough: *iterative decoding*.

(Iterative decoding is not a new idea, though: most of the key ideas were already present in Gallager’s work on low-density parity-check codes [2].)

With respect to Viterbi decoding, a code is most naturally described by means of a trellis diagram. The main thesis of the present paper is that, with respect to iterative decoding, the natural way of describing a code is by means of a Tanner graph [3], which may be viewed as a generalized trellis. More precisely, it is the “time axis” of a trellis that is generalized to a Tanner graph.

Trellises yield Tanner graphs of the type shown in Fig. 1; in particular, the graph has no cycles. The complexity reduction in turbo codes (and low-density parity-check codes, and many new codes to be discovered) comes from allowing Tanner graphs with cycles, cf. Fig. 2.

II. DECODING

Both Viterbi decoding and BCJR decoding [4] are easily generalized to arbitrary Tanner graphs *without cycles*, where these algorithms are still optimal (in the same sense as for trellises). The basic idea of iterative decoding is simply to apply these algorithms even to Tanner graphs *with cycles*, ignoring the fact that the algorithms are no longer optimal. The empirical success of turbo coding (as well as our own experiments with other types of codes) confirm the validity of this approach.

Of course, analytical understanding of the decoder operation is also desirable. Our main result here applies to “cycle codes” (a subclass of low-density parity-check codes): we give a complete algebraic characterization of all error patterns that are corrected by the generalized Viterbi algorithm after infinitely many iterations.

III. REALIZATION THEORY ON GENERAL GRAPHS

Much recent work was devoted to finding, and bounding the size of, the “smallest” trellis for a given code. This problem is significantly generalized by considering general Tanner graphs.

In the traditional setting, the only degree of freedom (for a given code) was the ordering of the “time axis”. For a given ordering, every linear code has a well-defined unique minimal trellis, and every other trellis for the same code may be collapsed to the minimal trellis by state merging.

In our more general setting, the “time axis” need not be ordered, but may be an arbitrary Tanner graph. Even for a fixed

Tanner graph, there is, in general, no unique minimal trellis. (The simplest example are tail-biting trellises.) Nevertheless, bounds on the “size” of the realization may be obtained from the (“abstract”) state spaces of the code.

IV. A PRIORI PROBABILITIES

Our careful derivation of the two basic iterative decoding algorithms clarifies, in particular, what a priori distributions are admissible and how they are properly dealt with. As it turns out, these distributions are closely related to Markov Random fields.

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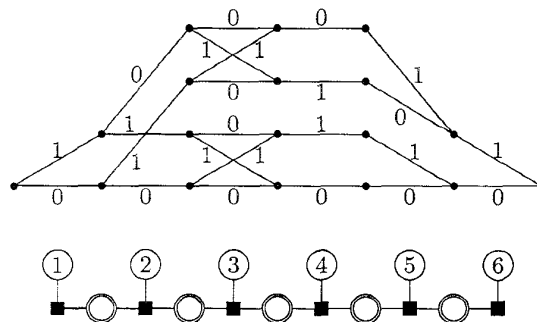


Figure 1: A trellis (top) and its Tanner graph (bottom).

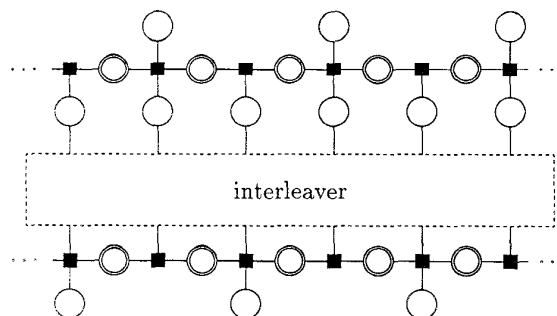


Figure 2: The Tanner graph of turbo coding [1].